

The exact Poincaré constant of the two-particle beta-Ginibre ensemble

Candidate exact solved-subcase packet

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Status: candidate exact solved subcase, likely valid pending expert review.

Abstract

Chafaï and Lehec ask for the Poincaré and logarithmic Sobolev constants of the beta-Ginibre ensemble. We determine the optimal Poincaré constant for two particles and every repulsion exponent $\beta \geq 0$. For the law on $\mathbb{C}^2$ with density proportional to

$$e^{-a(|z_1|^2+|z_2|^2)}|z_1 - z_2|^\beta, \quad a > 0,$$

the spectral gap is

$$\lambda_{a,\beta} = a(\sqrt{\beta^2 + 4} - \beta),$$

and the optimal Poincaré constant is its reciprocal, $(\sqrt{\beta^2 + 4} + \beta)/(4a)$. The proof factors center of mass from relative position, then diagonalizes the relative generator by angular Fourier modes and generalized Laguerre polynomials. For the normalization of arXiv:1805.00708 and two particles, $a = 2$.

1 Source question and the result

The source is D. Chafaï and J. Lehec, *On Poincaré and logarithmic Sobolev inequalities for a class of singular Gibbs measures*, arXiv:1805.00708 [1]. In its comments and open questions it considers the beta-Ginibre density

$$z \in \mathbb{C}^n \longmapsto e^{-n \sum_{j=1}^n |z_j|^2} \prod_{j < k} |z_j - z_k|^\beta$$

The exact source sentence is: “it is not clear at all what are the Poincaré and log-Sobolev constants of the Ginibre ensemble.”

and involve polynomials goes back at least to Calogero [17], more than twenty five years before Lassalle!

The factorization phenomenon captured by Lemma 2.6 which is behind the optimality provided by Theorem 1.3 reminds some kind of concentration-compactness related to continuous spins systems as in [21] and [58] for instance. The factorization Lemma 2.6 remains valid for other ensembles such as the Beta-Ginibre ensemble with density proportional to

$$z \in \mathbb{C}^n \mapsto e^{-n \sum_{k=1}^n |z_k|^2} \prod_{j < k} |z_j - z_k|^\beta, \quad (1.19)$$

see [24, Remark 5.4] for the case $n = \beta = 2$. However, in contrast with the Beta-Hermite ensemble, the interaction term is not convex in the complex case, and it is not clear at all what are the Poincaré and log-Sobolev constants of the Ginibre ensemble. See [24] for an upper bound and further discussions on the associated dynamics.

2. USEFUL OR BEAUTIFUL FACTS

2.1. Random matrices, GUE, and beta Hermite ensemble. The following result from random matrix theory goes back to Dyson, see [37, 43, 71, 74]

Figure 1: The beta-Ginibre question in arXiv:1805.00708, PDF page 8.

We solve the Poincaré part exactly when $n = 2$. It is useful to keep the quadratic confinement coefficient general. Let $\mu_{a,\beta}$ be the probability measure on $\mathbb{C}^2 \cong \mathbb{R}^4$ with density

$$\frac{d\mu_{a,\beta}}{dz_1 dz_2} = \frac{1}{Z_{a,\beta}} e^{-a(|z_1|^2 + |z_2|^2)} |z_1 - z_2|^\beta, \quad a > 0, \quad \beta \geq 0. \quad (1)$$

Gradients below are real Euclidean gradients on the two copies of $\mathbb{C}$.

Theorem 1 (Exact two-particle constant). *For every real f in the weighted Sobolev space $H^1(\mu_{a,\beta})$,*

$$\text{Var}_{\mu_{a,\beta}}(f) \leq \frac{\sqrt{\beta^2 + 4} + \beta}{4a} \int_{\mathbb{C}^2} (|\nabla_{z_1} f|^2 + |\nabla_{z_2} f|^2) d\mu_{a,\beta}. \quad (2)$$

The constant is optimal. Equivalently, the spectral gap is

$$\lambda_{a,\beta} = a(\sqrt{\beta^2 + 4} - \beta). \quad (3)$$

If $\beta > 0$, put

$$\alpha_1 = \frac{\sqrt{\beta^2 + 4} - \beta}{2}.$$

Apart from additive constants, equality is attained precisely by the real span of

$$|z_1 - z_2|^{\alpha_1 - 1} \Re(z_1 - z_2), \quad |z_1 - z_2|^{\alpha_1 - 1} \Im(z_1 - z_2).$$

For $\beta = 0$, all affine real functions attain equality.

Corollary 2 (Source normalization). *For the two-particle beta-Ginibre law in arXiv:1805.00708, $a = n = 2$, so*

$$C_P = \frac{\sqrt{\beta^2 + 4} + \beta}{8}, \quad \lambda = 2(\sqrt{\beta^2 + 4} - \beta).$$

For the classical complex Ginibre exponent $\beta = 2$ this gives

$$C_P = \frac{1 + \sqrt{2}}{4}, \quad \lambda = 4(\sqrt{2} - 1).$$

2 Proof intuition

Use orthonormal center and difference coordinates

$$u = \frac{z_1 + z_2}{\sqrt{2}}, \quad v = \frac{z_1 - z_2}{\sqrt{2}}.$$

The density then factors. The center u is an ordinary planar Gaussian, while the relative coordinate v has density $|v|^\beta e^{-a|v|^2}$. The only non-Gaussian issue is therefore a two-dimensional radial oscillator with a singular drift at the origin.

In polar coordinates, separate the angular mode $e^{im\theta}$. A trial factor r^α cancels the inverse-square term exactly when $\alpha(\alpha + \beta) = m^2$. After the change $\rho = ar^2$, the remaining equation is the generalized Laguerre equation. This gives the whole spectrum explicitly. The first excitation is not radial: it is the $|m| = 1$ angular mode, and its noninteger radial power is what makes the optimal constant larger than the Gaussian center-of-mass constant.

3 Factorization

The map $(z_1, z_2) \mapsto (u, v)$ is orthogonal on $\mathbb{R}^4$, and

$$|z_1|^2 + |z_2|^2 = |u|^2 + |v|^2, \quad |z_1 - z_2|^\beta = 2^{\beta/2} |v|^\beta.$$

The harmless power of 2 is absorbed by normalization. Hence

$$\mu_{a,\beta} = \gamma_a \otimes \nu_{a,\beta}, \tag{4}$$

where

$$d\gamma_a(u) = \frac{a}{\pi} e^{-a|u|^2} du, \quad d\nu_{a,\beta}(v) = c_{a,\beta} |v|^\beta e^{-a|v|^2} dv.$$

The coordinate change also preserves the squared gradient. The planar Gaussian factor has spectral gap $2a$. It remains to compute the gap of $\nu_{a,\beta}$.

4 The complete relative spectrum

In polar coordinates $v = (r, \theta)$, the relative measure is proportional to

$$r^{\beta+1} e^{-ar^2} dr d\theta.$$

The self-adjoint generator associated with the closure of the Dirichlet form $\int |\nabla f|^2 d\nu_{a,\beta}$ is the Friedrichs realization of

$$\mathcal{L} = \partial_r^2 + \left(\frac{\beta+1}{r} - 2ar \right) \partial_r + \frac{1}{r^2} \partial_\theta^2. \tag{5}$$

Initially this identity may be read on smooth functions supported away from $r = 0$; the form closure supplies the natural boundary condition there.

For $m \in \mathbb{Z}$, define

$$\alpha_m = \frac{\sqrt{\beta^2 + 4m^2} - \beta}{2}, \quad q_m = \alpha_m + \frac{\beta}{2}. \tag{6}$$

Thus $\alpha_m \geq 0$ and $\alpha_m(\alpha_m + \beta) = m^2$.

Proposition 3 (Relative spectral decomposition). *For $m \in \mathbb{Z}$ and $k \in \mathbb{N}_0$, let*

$$\Phi_{m,k}(r, \theta) = r^{\alpha_m} L_k^{(q_m)}(ar^2) e^{im\theta}, \tag{7}$$

where $L_k^{(q)}$ is the generalized Laguerre polynomial. These functions form a complete orthogonal eigenfamily in the complexification of $L^2(\nu_{a,\beta})$, and

$$-\mathcal{L}\Phi_{m,k} = (4ak + 2a\alpha_m)\Phi_{m,k}. \quad (8)$$

Consequently

$$\text{gap}(\nu_{a,\beta}) = a(\sqrt{\beta^2 + 4} - \beta).$$

Proof. Fix an angular mode $e^{im\theta}$ and write $R(r) = r^\alpha y(\rho)$ with $\rho = ar^2$. The radial operator in that mode is

$$\mathcal{L}_m R = R'' + \left(\frac{\beta + 1}{r} - 2ar \right) R' - \frac{m^2}{r^2} R.$$

A direct differentiation, using $\alpha(\alpha + \beta) = m^2$, gives

$$\mathcal{L}_m(r^\alpha y(ar^2)) = 4ar^\alpha \left[\rho y'' + (\alpha + \frac{\beta}{2} + 1 - \rho)y' - \frac{\alpha}{2}y \right]. \quad (9)$$

The Laguerre equation

$$\rho y'' + (q + 1 - \rho)y' + ky = 0$$

with $q = \alpha + \beta/2$ now proves (8).

For completeness, Fourier series decompose the angular variable. In the m th sector, multiplication by r^{α_m} and the substitution $\rho = ar^2$ identify the radial L^2 norm, up to a constant, with

$$\int_0^\infty |y(\rho)|^2 \rho^{q_m} e^{-\rho} d\rho.$$

Since $q_m > -1$, the generalized Laguerre polynomials are complete in this space. The positive root α_m is the finite-energy root selected by the Friedrichs form. The other root is inadmissible: for $m \neq 0$ it is not square integrable, and for $m = 0$ its nonconstant branch has infinite energy. Thus the displayed family exhausts the spectrum.

The constant is $m = 0, k = 0$. The nonzero radial eigenvalues begin at $4a$. For $m \neq 0$, α_m increases with $|m|$, so the first angular eigenvalue is

$$2a\alpha_1 = a(\sqrt{\beta^2 + 4} - \beta).$$

Because $\sqrt{\beta^2 + 4} - \beta \leq 2$, this is at most $2a$, and hence is strictly below $4a$. It is the relative spectral gap. $\square$

5 Proof of Theorem 1

By (4), the full generator is the tensor sum of the center generator and the relative generator. Its spectrum consists of sums of factor eigenvalues, so its gap is the smaller of the factor gaps:

$$\min \left\{ 2a, a(\sqrt{\beta^2 + 4} - \beta) \right\} = a(\sqrt{\beta^2 + 4} - \beta).$$

The reciprocal of this gap is

$$\frac{1}{a(\sqrt{\beta^2 + 4} - \beta)} = \frac{\sqrt{\beta^2 + 4} + \beta}{4a},$$

which proves (2) with its optimal constant.

For $\beta > 0$, the gap is strictly smaller than the center gap. The real $m = \pm 1, k = 0$ eigenfunctions in the relative coordinate are $r^{\alpha_1} \cos \theta$ and $r^{\alpha_1} \sin \theta$. Rewriting them in terms of $z_1 - z_2$ gives the equality cases stated in the theorem. They lie in the form domain: near zero their energy density is integrable, and cutoff approximations have vanishing extra energy. When $\beta = 0$, both factors are ordinary planar Gaussians with the same gap $2a$, and all affine real functions are extremal. This completes the proof.

6 Checks, scope, and novelty status

The formula has the required Gaussian limit: at $\beta = 0$ it gives gap $2a$. It also has the strong-repulsion asymptotic

$$\lambda_{a,\beta} \sim \frac{2a}{\beta}, \quad C_P \sim \frac{\beta}{2a}.$$

An accompanying exact symbolic script verifies the radial conjugation identity, the Laguerre residual, the ordering of the candidate eigenvalues, and the specializations $\beta = 0$ and $\beta = 2$.

This is a complete answer for the Poincaré constant of the two-particle family, hence a solved subcase of the source’s arbitrary- n question. It does not determine the constant for arbitrary particle number or the logarithmic Sobolev constant. A bounded novelty search through the run indexes, exact-title searches, beta-Ginibre functional-inequality searches, and searches for the resulting closed formula found no prior statement of this exact two-particle constant. Chafaï, Bolley, and Fontbona prove existence of a Poincaré inequality and discuss non-explicit bounds for the planar Coulomb gas [2]; that result is consistent with, but does not contain, the spectral calculation above. This novelty check is not an exhaustive bibliographic certification.

For human review, the two points deserving the closest check are the Friedrichs-domain selection at the collision point and the claim that the Fourier–Laguerre family exhausts the relative spectrum. Both are addressed in Proposition 3; an independent symbolic residual and Rayleigh-quotient check is included with the packet.

References

- [1] D. Chafaï and J. Lehec, *On Poincaré and logarithmic Sobolev inequalities for a class of singular Gibbs measures*, arXiv:1805.00708 (2018), <https://arxiv.org/abs/1805.00708>.
- [2] F. Bolley, D. Chafaï, and J. Fontbona, *Dynamics of a planar Coulomb gas*, arXiv:1706.08776 (2017), <https://arxiv.org/abs/1706.08776>.